Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/09/2005
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Project Title:	Measurement of Flow Field Distortion and Particle Equation of Motion in Turbulent Gas Flows		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Zip Code:	94305-3030	**Congressional District:**	14
Comments:			
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	1. The first task is to conduct global measurements of turbulence modification and preferential concentration in the homogeneous and isotropic turbulence chamber. The experiments will be conducted in the laboratory and in micro-gravity to examine the effect of mean particle wakes due to gravitational settling. 2. The second task is to develop a high resolution hybrid PIV/PTV system that can spatially resolve down to the smallest eddy in the flow, and conduct local measurements. The system must be capable of measuring the fluid velocity in the vicinity of particles, where large amounts of light are scattered. It must also be able to estimate the local energy dissipation field around particles. The present research aims at investigating the physical mechanism of turbulence modification by particles. For this purpose we have developed a turbulence chamber that creates the most basic type of turbulent flow, which is homogeneous and isotropic turbulence without mean flow. Global measurements are first needed to increase the understanding of the overall phenomenon. Turbulence modification experiments are needed in the laboratory where gravity exists and in micro-gravity on NASA’s KC-135 to evaluate the effects of the mean particle		

Task Description:	wakes. Preferential concentration of particles also must be examined. Second, local measurements of both phases in the vicinity of individual particles need to be made. A high resolution hybrid PIV/PTV system will enable us to calculate the aerodynamic force on the particles and investigate the particle equation of motion. 1. Gravity plays an important role in fluid-particle interaction in turbulent flows. Eliminating this parameter will give us a better understanding of this complex phenomenon. The turbulence chamber that has been developed is capable of creating turbulence with high Taylor microscale Reynolds numbers and no inflow or outflow. It is ideal for use in a microgravity environment where space is limited. The ideal case of homogeneous and isotropic turbulence will provide a database for modelers and a basis of comparison with ongoing direct numerical simulations (DNS). Such a chamber could also be used for other turbulence research being conducted in a microgravity environment. 2. The high resolution hybrid PIV/PTV system that has been developed enables us to calculate the aerodynamic force on the particles and investigate the particle equation of motion. The results will be used to test various models for turbulence modification in particle-laden flows. We believe it will be able to play a significant role in resolving controversy about the mechanisms and modeling of turbulence attenuation by particles. 3. Energy dissipation is one of the most important factors to clarify the mechanisms of turbulence attenuation. The sink of turbulence energy is generally due to ordinary fluid viscous dissipation and extra dissipation caused by local distortion of the flow field around particles. Previous experiments and simulations have not been capable of resolving this extra dissipation. The present experiment intends to quantify the dissipation caused by local velocity disturbances in the vicinity of the small particles in homogeneous and isotropic flow using high-resolution particle image velocimetry (PIV). The spatial resolution of our custom PIV system is approximately one seventh of the particle diameter and this allows us to estimate the local dissipation field around particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Global measurements of turbulence modification by particles in homogeneous and isotropic turbulence have been completed. Laboratory experiments indicate that the turbulent kinetic energy and viscous dissipation rate are both attenuated with increasing mass loading (up to about 40% at a mass loading of 0.23) of particles that are on the order of the Kolmogorov length scale. Clusters of particles added energy to the fluid via local jets at large scales and individual particles added energy through particle wakes at small scales. The fluid fluctuations in the vicinity of the fluctuating particles were quickly dissipated and acted as an extra sink of energy for the fluid. The micro-gravity experiments showed that the absence of particle potential energy loss and particle wakes caused greater levels of turbulence attenuation since there was no additional production due to the particles. The particles probably acted like a series of screens which hampered the evolution of the turbulence. Local measurements are being conducted with a high-resolution PIV/PTV system that has been developed to examine the particle equation of motion. Currently, PIV is measuring the flowfield distortion around individual particles. Strong energy dissipation is clearly observed in this region. In addition, particle tracking velocimetry (PTV) will track the particle motion to obtain particle acceleration information. Measurements will soon follow.
Bibliography Type:	Description: (Last Updated: 08/22/2007)
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